

Assignment: Discuss 4 Questions

1. Why Consensus Exists (centralized vs decentralized)
 - ⇒ Consensus exists because decentralized systems don't have a central authority to decide what information is correct. In centralized systems, a single organization controls the database and verifies transactions, so consensus is not necessary. However, in decentralized systems like blockchain, many independent nodes participate in the network, and consensus is required so that all nodes agree on which transactions are valid, which block is the latest, and what the correct global ledger state is.
 2. Safety, Liveness, Finality
 - ⇒ Safety, liveness, and finality are the three main properties of a consensus mechanism. Safety ensures that invalid or conflicting transactions are never accepted and that the system does not produce inconsistent results, such as double spending. Liveness guarantees that the system continues to operate and that valid transactions will eventually be processed and included in the blockchain. Finality means that once a transaction is confirmed, it becomes permanent and cannot be reversed, giving users confidence that the transaction is complete.
 3. Consensus in P2P Networks (visual topology)
 - ⇒ In a peer-to-peer network, there is no central server, and each node connects directly with other nodes. Every node maintains a copy of the blockchain and independently verifies transactions and blocks. Consensus allows these nodes to communicate, validate information, and eventually agree on a single shared version of the blockchain, even though messages may arrive at different times.
 4. Common Attack Models
 - ⇒ Common attack models target weaknesses in decentralized networks. A Sybil attack happens when an attacker creates many fake identities to influence the network. Double spending is an attempt to use the same digital asset more than once, and eclipse attacks isolate a node and feed it incorrect information. Consensus mechanisms are designed to reduce or prevent the impact of these attacks.
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